3blue 1brown

Exponential Weighted Average

Gradient Descent with Momentum

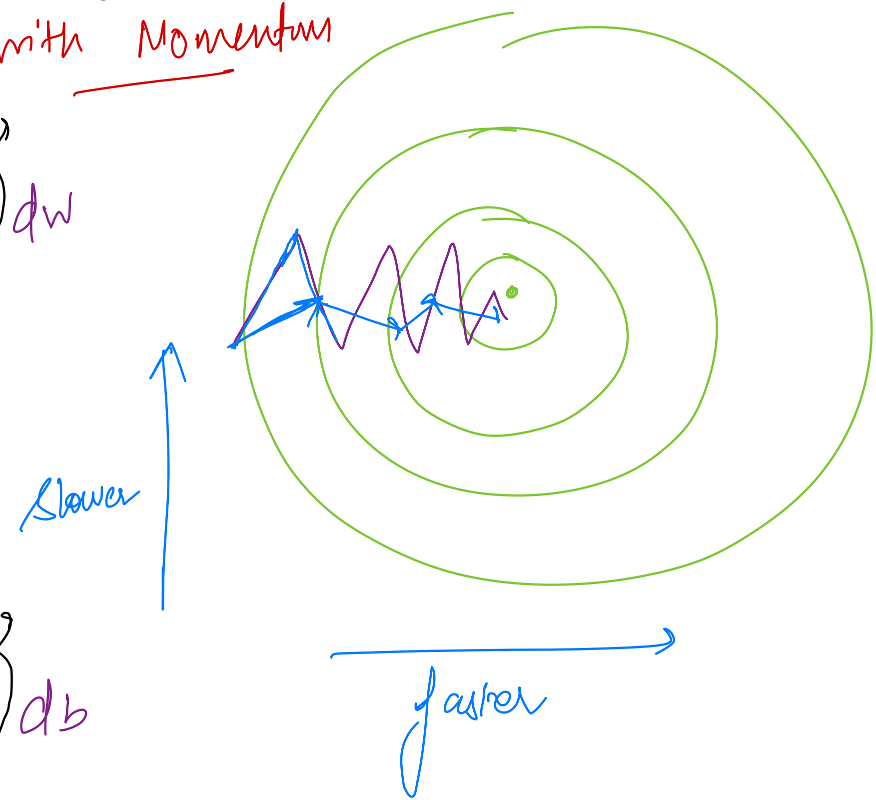
$$V_{dw} = \beta V_{dw} + \cancel{(1-\beta)}\, dw$$

Updating weights

$$W = W - \alpha V_{dw}$$

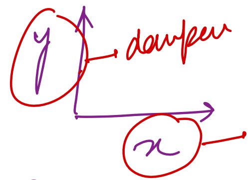
$$V_{db} = \beta V_{db} + \cancel{(1-\beta)}\, db$$

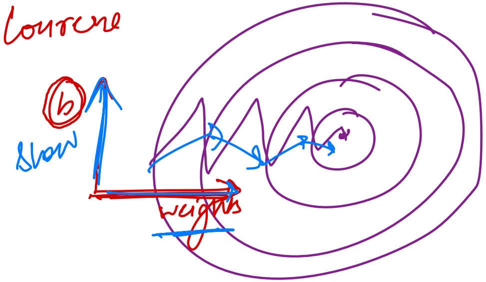
$$b = b - \alpha V_{db}$$



RMS Prop → Geoffery Hinton → Coursera

Root mean square





$V_{dw} = \beta (V_{dw}) + (1-\beta)\, dw$ → gradient descent with momentum

$S_{dw} = \beta (S_{dw}) + (1-\beta)\, dw^2$ → smaller

$S_{db} = \beta\, S_{db} + (1-\beta)\, db^2$

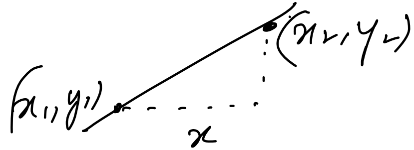
$W = W - \dfrac{\alpha\, dw}{\sqrt{S_{dw}}}$ → focus on increasing

make it small

learning to be faster in x direction →

$\Delta W \neq dW$

$W_2 - W_1$



$\Delta x = x_2 - x_1$

$\Delta y = y_2 - y_1$

$$b = b - \alpha \frac{db}{\sqrt{S_{db}}}$$

= small

big

learning small / slow

Adam Optimizer

$$W = W - \alpha \frac{V_{dW}}{\sqrt{S_{dW}}}$$

$$b = b - \alpha \frac{V_{db}}{\sqrt{S_{db}}}$$

0.01

Learning rate decay

new learning rate

$$\alpha = \frac{1}{1 + \delta_0 \times \text{epoch}} \times \alpha_0$$

old learning

rate (0.01)

$$\alpha = \frac{1}{1+(0.01)\times 100} \times 0.01$$

$$\Rightarrow \frac{0.01}{2} \Rightarrow \underline{\underline{0.005}}$$